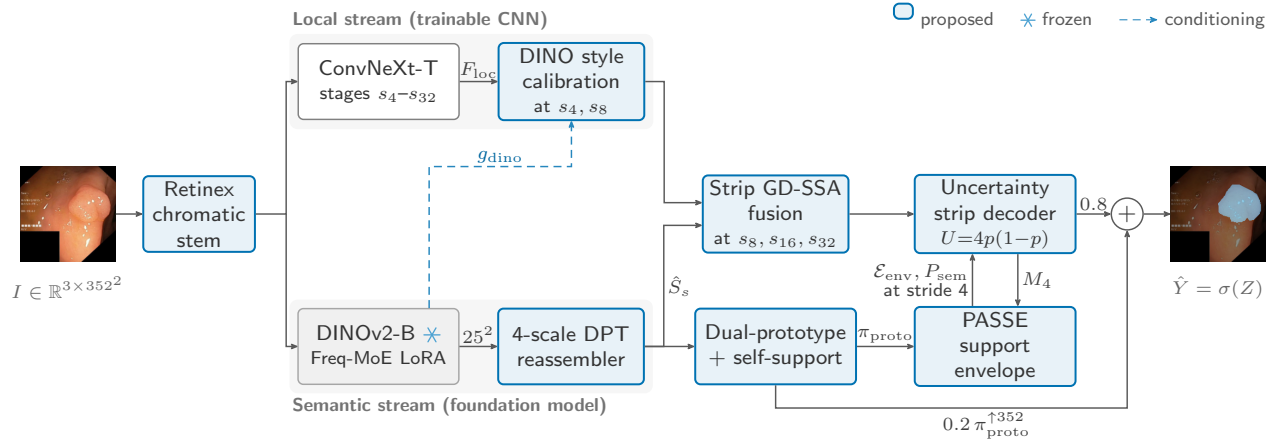

proposed
frozen
conditioning
Local stream (trainable CNN)
ConvNeXt-T
stages s_4–s_{32}
F_{loc}
DINO style calibration
at s_4, s_8
g_{dino}
Retinex chromatic stem
$I \in \mathbb{R}^{3\times 352^2}$
DINOv2-B
Freq-MoE LoRA
25^2
4-scale DPT reassembler
Semantic stream (foundation model)
$\hat{S}_s$
Strip GD-SSA fusion
at s_8, s_{16}, s_{32}
Dual-prototype + self-support
π_{proto}
$\mathcal{E}_{\mathrm{env}}, P_{\mathrm{sem}}$
at stride 4
M_4
PASSE support envelope
Uncertainty strip decoder
$U{=}4p(1{-}p)$
0.8
$0.2\,\pi_{\mathrm{proto}}^{\uparrow 352}$
$\hat{Y} = \sigma(Z)$